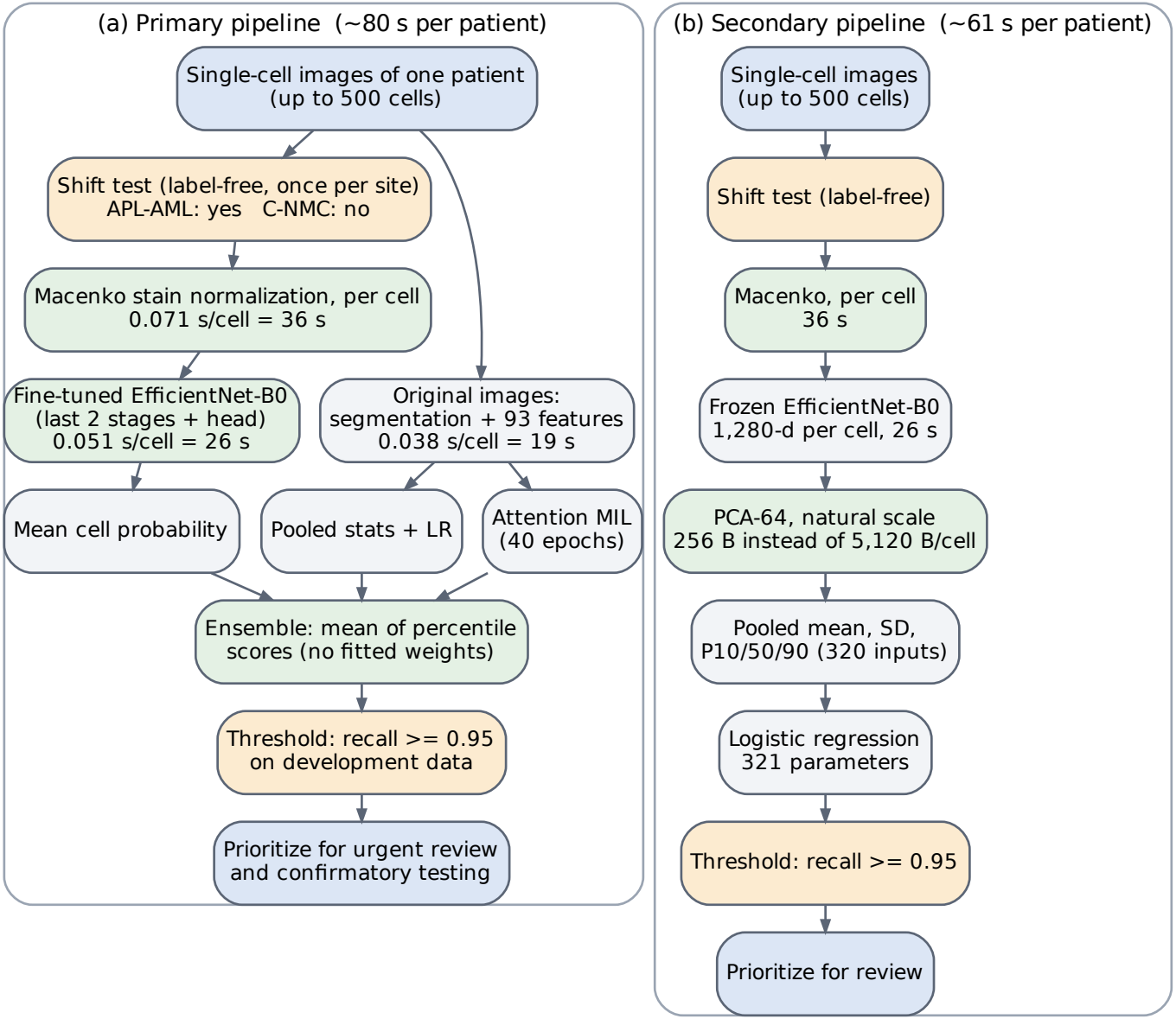
(a) Primary pipeline (~80 s per patient)
Single-cell images of one patient (up to 500 cells)
Shift test (label-free, once per site) APL-AML: yes C-NMC: no
Macenko stain normalization, per cell 0.071 s/cell = 36 s
Fine-tuned EfficientNet-B0 (last 2 stages + head) 0.051 s/cell = 26 s
Original images: segmentation + 93 features 0.038 s/cell = 19 s
Mean cell probability
Pooled stats + LR
Attention MIL (40 epochs)
Ensemble: mean of percentile scores (no fitted weights)
Threshold: recall >= 0.95 on development data
Prioritize for urgent review and confirmatory testing
(b) Secondary pipeline (~61 s per patient)
Single-cell images (up to 500 cells)
Shift test (label-free)
Macenko, per cell 36 s
Frozen EfficientNet-B0 1,280-d per cell, 26 s
PCA-64, natural scale 256 B instead of 5,120 B/cell
Pooled mean, SD, P10/50/90 (320 inputs)
Logistic regression 321 parameters
Threshold: recall >= 0.95
Prioritize for review